Chapter 11. Smoothing and Density Estimation

Histograms, kernel density estimation, bandwidth selection, multivariate KDE, confidence bands

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11.1 From histograms to kernels

The empirical distribution function estimates F well but has no derivative, so it gives no direct estimate of the density f. A histogram does, but it depends on both the bin width and the bin origin. Averaging histograms over shifted origins gives a smooth estimate: the kernel estimator.

Definition (Kernel density estimator)

For a kernel K and bandwidth $h > 0$, $\hat{f}(x) = \dfrac{1}{nh}\sum_{i=1}^{n} K\Big(\dfrac{x - X_i}{h}\Big)$.

Standard kernels satisfy $K \geq 0$, $\int K = 1$, $K(-u) = K(u)$, and $\sigma_K^2 = \int u^2 K < \infty$. Examples are the Gaussian ϕ, the Epanechnikov kernel $\frac{3}{4}(1-u^2)_+$ and the uniform kernel $\frac{1}{2}I(|u| \leq 1)$. Then $\hat{f} \geq 0$ and $\int \hat{f} = 1$ (substitute $u = (x - X_i)/h$), so $\hat{f}$ is a density. The choice of h matters far more than the choice of K.

Example 11.1

Data $\{0, 1, 3\}$, Gaussian kernel, $h = 1$: $\hat{f}(1) = \frac{1}{3}[\phi(1)+\phi(0)+\phi(2)] = \frac{1}{3}(0.242+0.399+0.054) = 0.232$. The point 3, two bandwidths away, contributes little.

11.2 Bias, variance and bandwidth

Motivation. A kernel density estimate is controlled by its bandwidth: small bandwidths give noisy estimates, large ones oversmooth. Computing bias and variance shows how the optimal bandwidth shrinks with n and gives the $n^{-4/5}$ rate.

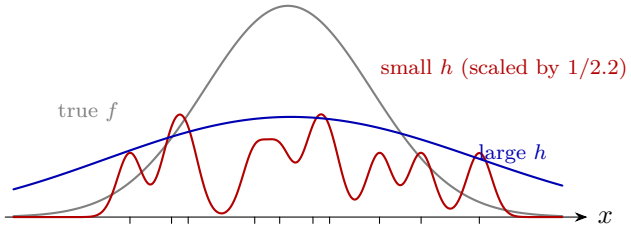


Figure. *Kernel density estimates from the same ten points (ticks). A small bandwidth gives a spiky estimate with large variance (its vertical scale is shrunk here to fit); a large bandwidth flattens the peak, a bias. The optimal $h \propto n^{-1/5}$ balances the two.*

Theorem 11.1

Let f be twice continuously differentiable near x with $f(x) > 0$. As $n \to \infty$, $h \to 0$ and $nh \to \infty$,

$$\operatorname{bias}\hat{f}(x) = \frac{\sigma_K^2 f''(x)}{2}h^2 + o(h^2), \qquad \operatorname{Var}\hat{f}(x) = \frac{f(x)R(K)}{nh} + o\Big(\frac{1}{nh}\Big), \qquad R(K) = \int K^2.$$

Proof [Supplement]. $E\hat{f}(x) = \int \frac{1}{h}K(\frac{x-t}{h})f(t)\,dt = \int K(u)f(x-hu)\,du$. Expand $f(x-hu) = f(x) - huf'(x) + \frac{h^2u^2}{2}f''(x) + o(h^2u^2)$, use $\int uK = 0$ and $\int u^2K = \sigma_K^2$, and justify the remainder by dominated convergence. For the variance, $\operatorname{Var}\hat{f}(x) = \frac{1}{n}\operatorname{Var}\big(\frac{1}{h}K(\frac{x-X}{h})\big)$. Here $E\frac{1}{h^2}K^2(\frac{x-X}{h}) = \frac{1}{h}\int K^2(u)f(x-hu)\,du = \frac{f(x)R(K)}{h} + O(1)$, while the squared mean is $O(1)$. ∎

This is the **bias–variance trade-off**: a small h gives low bias and high variance (a wiggly estimate), and a large h the reverse. Integrating,

$$\mathrm{MISE}(\hat f) = E\int (\hat f - f)^2 \approx \frac{\sigma_K^4 R(f'')}{4} h^4 + \frac{R(K)}{nh}, \qquad R(f'') = \int (f'')^2,$$

minimized at

$$h^* = \Big(\frac{R(K)}{\sigma_K^4 R(f'')\, n}\Big)^{1/5} \asymp n^{-1/5}, \qquad \mathrm{MISE}(h^*) \asymp n^{-4/5}. \tag{11.1}$$

(Set the derivative $\sigma_K^4 R(f'')h^3 - R(K)/(nh^2)$ to zero.) The rate $n^{-4/5}$ is slower than the parametric rate n^{-1}: this is the price of not assuming a shape.

Choosing h. The optimal h^* contains the unknown $R(f'')$.

- **Normal reference.** If $f = N(\mu, \sigma^2)$ then $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$, so with the Gaussian kernel $h = 1.06\,\hat\sigma n^{-1/5}$, where $\hat\sigma = \min(s, \mathrm{IQR}/1.34)$.
- **Silverman's rule of thumb:** $h = 0.9\,\hat\sigma n^{-1/5}$, slightly smaller to protect against multimodality.
- **Plug-in (Sheather–Jones):** estimate $R(f'')$ by a pilot kernel estimate of f'' and substitute into (11.1). This is generally the recommended choice; cross-validation is an alternative.

Proof of the normal value of $R(f'')$. For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi^2 = \frac{1}{2\sqrt{\pi}} \times$ the $N(0, \frac12)$ density. So $R(\phi'') = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2$ with $Z \sim N(0, \frac12)$, which is $\frac{1}{2\sqrt{\pi}}(\frac34 - 1 + 1) = \frac{3}{8\sqrt{\pi}}$. Scaling gives the factor σ^{-5}. With $R(\phi) = \frac{1}{2\sqrt{\pi}}$ and $\sigma_K = 1$, (11.1) gives $h^* = (4/3)^{1/5}\sigma n^{-1/5} = 1.06\sigma n^{-1/5}$. ■

11.3 Multivariate density estimation

For $\mathbf{X}_i \in \mathbb{R}^p$, $\hat f(\mathbf{x}) = \frac1n \sum_i K_H(\mathbf{x} - \mathbf{X}_i)$ with $K_H(\mathbf{x}) = |H|^{-1/2} K(H^{-1/2}\mathbf{x})$ and a bandwidth matrix H (e.g. $H = \mathrm{diag}(h_1^2, \dots, h_p^2)$, or a plug-in estimate of a full matrix that follows the shape of the data). With a product kernel and equal widths h,

$$\mathrm{MSE}\,\hat f(\mathbf{x}) \approx \Big[\frac{\sigma_K^2 h^2}{2} \sum_j \partial_{jj} f(\mathbf{x})\Big]^2 + \frac{f(\mathbf{x}) R(K)^p}{nh^p},$$

so $h^* \asymp n^{-1/(p+4)}$ and $\mathrm{MSE} \asymp n^{-4/(p+4)}$. This is the **curse of dimensionality**: in $p = 10$ dimensions, a window of side 0.1 in the unit cube contains on average $10^{-10}n$ points. Kernel density estimation is practical in one or two dimensions, occasionally three.

11.4 Confidence intervals and bands

Since bias $\hat f(x)$ and se $\hat f(x)$ are of the same order at h^*, intervals are stated for the smoothed density $f_h(x) = E\hat f(x)$ rather than for $f(x)$.

- **Pointwise, plug-in:** by the CLT, $\sqrt{nh}(\hat f(x) - f_h(x)) \xrightarrow{d} N(0, f(x)R(K))$, so $f_h(x) \in \hat f(x) \pm z_{\alpha/2}\sqrt{\hat f(x) R(K)/(nh)}$.
- **Pointwise, bootstrap:** resample the data, recompute $\hat f^*$ on a grid with the same h, and use $\hat f \pm z_{\alpha/2}\widehat{\mathrm{se}}^*$, the basic interval $(2\hat f - Q^*_{1-\alpha/2}, 2\hat f - Q^*_{\alpha/2})$, or the symmetric interval $\hat f \pm Q_{1-\alpha}(|\hat f^* - \hat f|)$.

- **Simultaneous band:** with $\Delta^* = \max_{x \in \text{grid}} |\hat{f}^*(x) - \hat{f}(x)|$ and its $1-\alpha$ quantile q^*, the band $\hat{f}(x) \pm q^*$ covers f_h on the whole grid with asymptotic probability $1-\alpha$. It is much wider than the pointwise intervals.

Negative lower limits can be truncated at 0 without changing coverage.

The quantity $(\hat{f}(x) - f_h(x))/\sqrt{\hat{f}(x)R(K)/(nh)}$ is asymptotically pivotal, which suggests a bootstrap-t interval with $t^* = (\hat{f}^* - \hat{f})/\sqrt{\hat{f}^* R(K)/(nh)}$. In practice it is unstable: where $\hat{f}^*(x) \approx 0$ the denominator vanishes, t^* explodes, and the interval becomes absurdly wide. The same problem rules out bands built from studentized maxima. In practice, pointwise intervals on a grid are reported more often than simultaneous bands. Some authors call such a collection a "confidence band", so check which is meant.